

Saveetha Institute of Medical and Technical Sciences

develops an eco-friendly, low-cost bioplastic from water hyacinth to protect environment

A low-cost bioplastic made of water hyacinth has been developed by a team led by Dr.Ashok kumar Veeramuthu, Head of the Centre for Waste Management and Renewable Energy of Saveetha Insitute of Medical and Technical Sciences, Chennai. The development is expected to help curb the scourge of the invasive aquatic weed and also address plastic pollution.

Dr.Ashokkumar's team successfully converted water hyacinth into high-quality polyhydroxyalkanoates (PHA), a fully biodegradable bioplastic poised to replace conventional single-use plastics. Explaining the three-step process, Dr.Ashokkumar told that the water hyacinth is collected and dried, then natural polymers/polysaccharides are extracted from the dried biomass.

Finally, through the biochemical conversion process, the polymers are transformed into bioplastics. "This low-cost method would tackle the two major environmental crises: the rampant spread of water hyacinth choking the water bodies and the scourge of persistent plastic pollution. Water hyacinth isn't just a weed; it's an ecological and economic disaster. It forms dense mats that clog lakes, rivers, and vital irrigation canals across India, blocking sunlight, depleting oxygen levels, decimating fish populations, hindering navigation, and exacerbating flooding. This directly harms biodiversity, fisheries, agriculture, and local livelihoods dependent on these water bodies. We transform this pervasive environmental head-ache into a valuable, sustainable resource-high-grade bioplastic," explained Dr.Ashokkumar.

This weed-based bioplastic naturally decomposes without polluting the environment, unlike conventional plastic that lasts for centuries. Developed from invasive water hyacinth using simple and affordable methods, the innovation offers a sustainable solution for waste management. The eco-friendly material can be used for plates, cutlery, bags and agricultural films. "The process is low-cost and uses locally available materials," said Dr. Ashokkumar. Saveetha Institute of Medical and Technical Sciences is now scaling up the project to reduce plastic pollution, clean water bodies and promote a greener future.